

Curtis's Ch9 final project

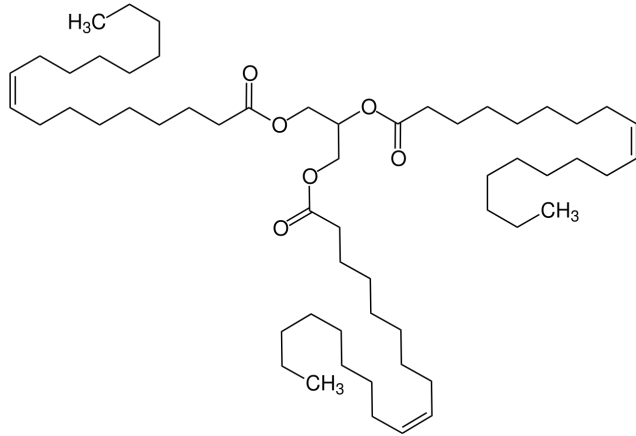
Production of biodiesel (Fatty Acid Ethyl Esters) from Used cooking oil and Ethanol

Background

- Biodiesel is a renewable diesel alternative produced by transesterifying oils or fats with a short-chain alcohol.
- Most commercial biodiesel is produced from vegetable oils or waste cooking oil using **methanol**, due to its fast reaction rate and easy product separation.
- **Ethanol** is a greener alternative because it can be produced from renewable biomass and is less toxic than methanol.
- However, ethanol is rarely used industrially because it leads to slower reaction kinetics, greater soap formation, and more difficult separation of biodiesel and glycerol.
- This project investigates the feasibility of producing biodiesel from **used cooking oil and ethanol**, evaluating whether the environmental benefits of ethanol can outweigh its processing challenges.

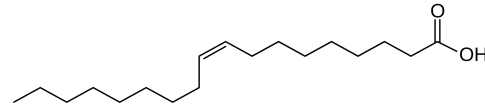
Composition of Used Cooking oil

Triglycerides 80-90%

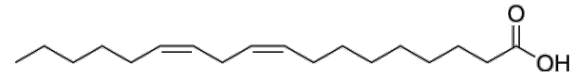


Triolein

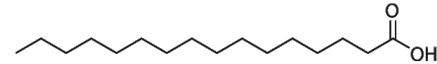
Free Fatty Acids 1-15%



Oleic Acid 30-45%

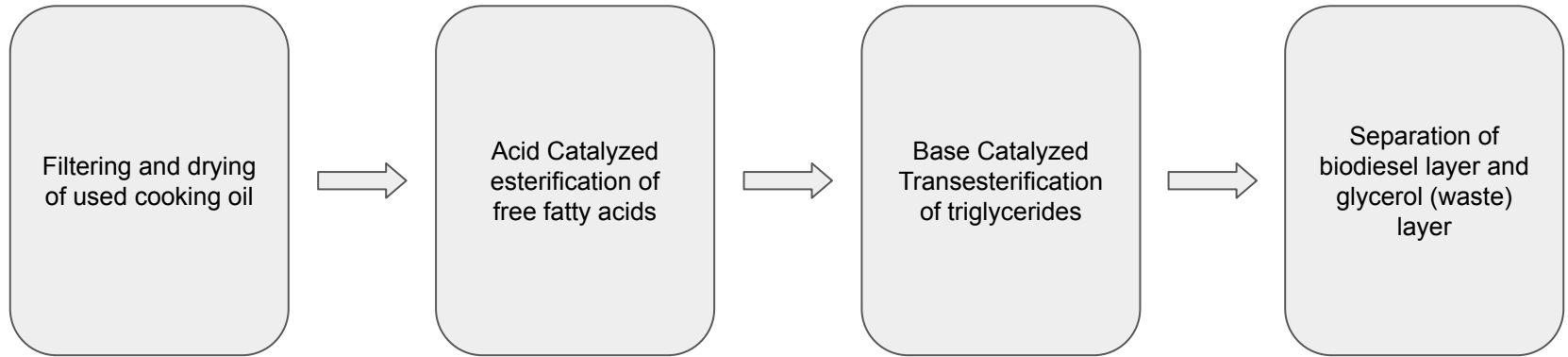


Linoleic Acid 20-30%



Palmitic Acid 10-15%

Synthesis steps



Titration of oil to determine free fatty acid (FFA) content



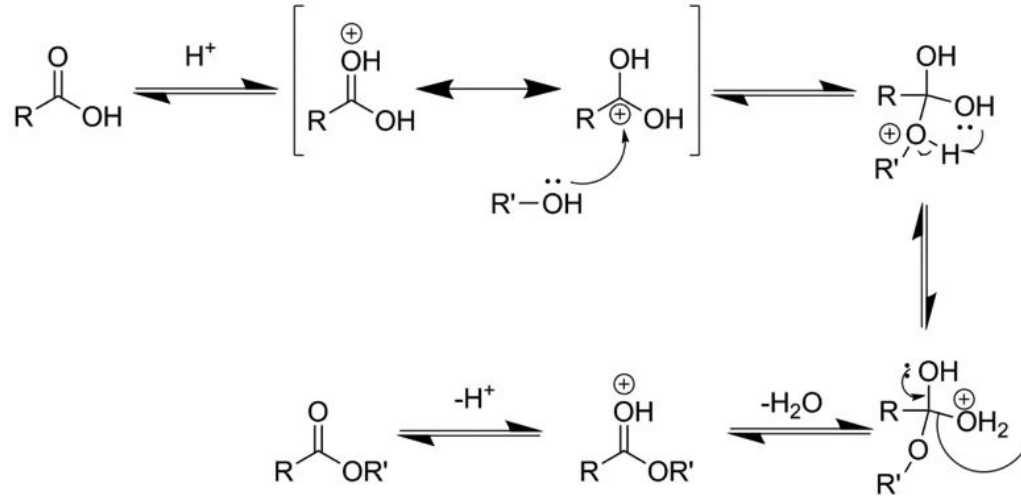
Using 0.01M of NaOH dissolved in isopropanol, and 5mL of oil dissolved in 15mL of isopropanol

2.2mL of base solution turned the solution light pink, which is 0.0044M FFA. Or if we assume it is Oleic acid, it is 0.12% by weight of oil.

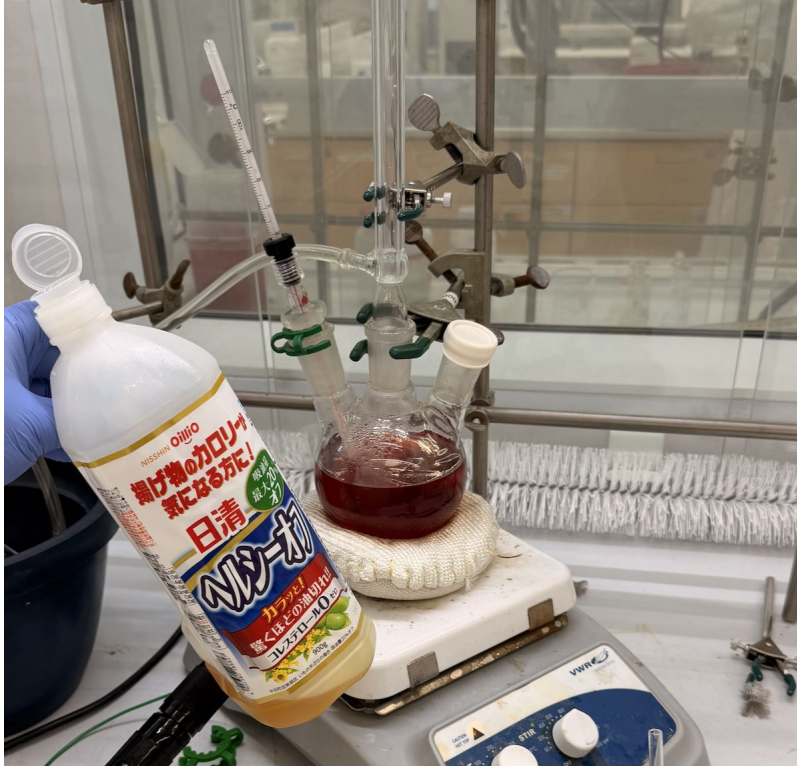
Acid Catalyzed Esterification of FFAs

Free fatty acid + Ethanol $\rightarrow$ Fatty Acid Ethyl Ester + Water

Conditions:
5g H₂SO₄ to 1g FFA
19.8:1 molar ratio of
ethanol to FFA



Transesterification of Used Cooking Oil



Recipe:

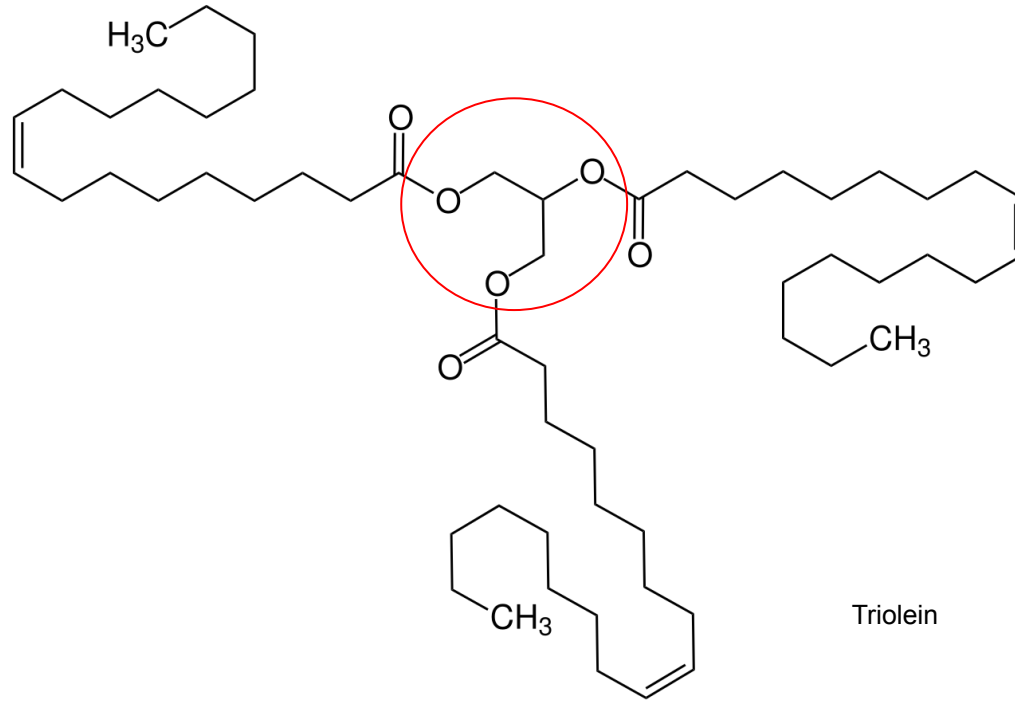
250mL Oil

120mL Ethanol (8 mol ratio EtOH:Oil)

2.5g KOH powder (5 wt% of oil)

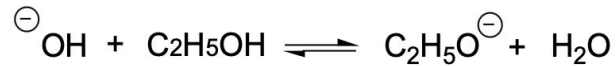
Stir and boil at 78°C for 70 minutes under full reflux

Transesterification of Used Cooking Oil



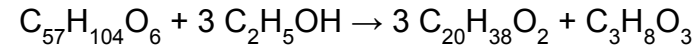
Transesterification of Used Cooking Oil

Epoxide formation

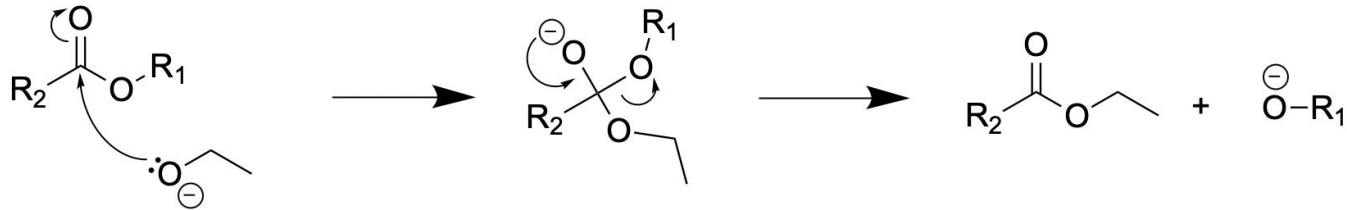


Balanced equation

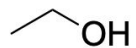
Triolein + 3 Ethanol $\rightarrow$ 3 Ethyl Oleate + Glycerol



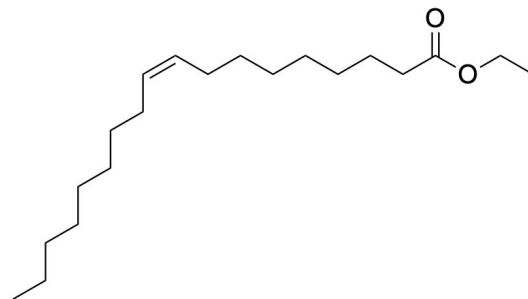
2-step Nucleophilic Acyl Substitution



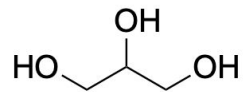
Separation



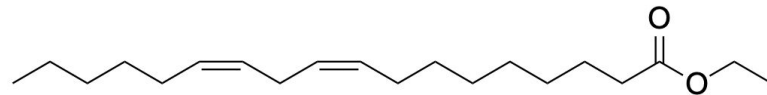
Ethanol



Ethyl Oleate



Glycerol



Ethyl Linoleate

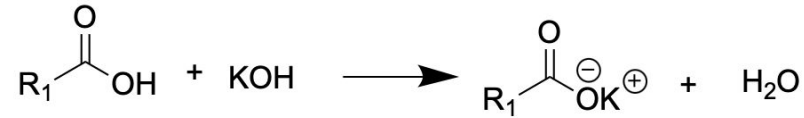
Ethanol recovery through distillation



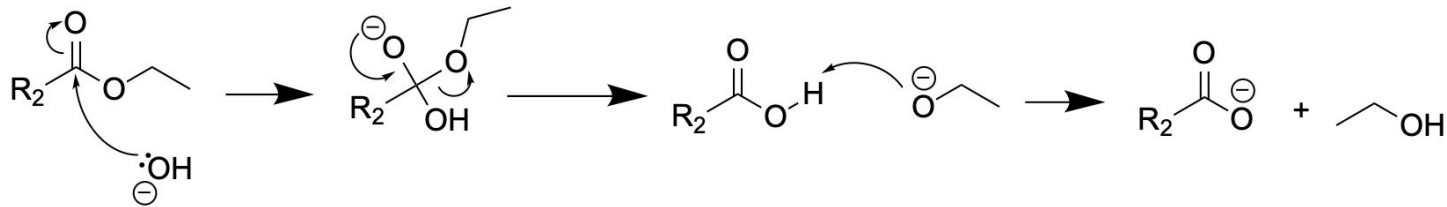
At 100% conversion and 100% triglyceride content, 45mL ethanol would be used, leaving 75mL. Flask distillation was used to remove ~90mL of ethanol until bubbling was reduced.

Saponification

Acid/Base reaction with free fatty acids



Base-catalyzed hydrolysis of ester



Carboxylate salt acts as a surfactant as it is very amphiphilic

Glycerol addition to encourage separation

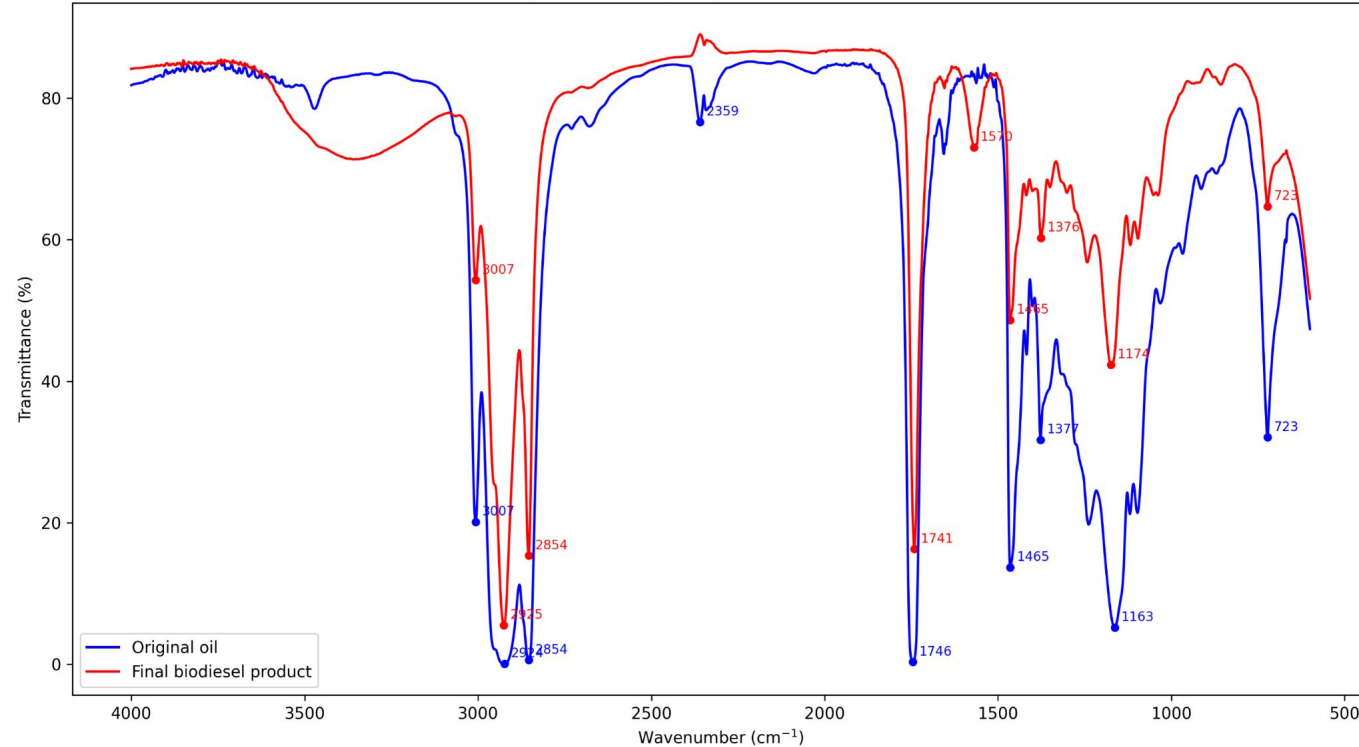


The addition of glycerol can encourage/induce phase separation it gives the system a larger denser polar phase which glycerol, soaps and salts can catalyst can preferentially partition into. Note the purple line is how much glycerol was added initially.

For 100% conversion of oil, 19mL of glycerol would have been produced.

Results

IR Spectra Overlay: Original Oil vs Final Product



Wavenumber cm⁻¹

Assignment

3100-3300

O-H stretch peak from Ethanol

3007

Sp² C-H stretch from unsaturated chains

2925, 2854

Sp³ C-H stretch

1746, 1741

C=O stretch from esters and triglycerides

1570

Carboxylate COO⁻ stretch

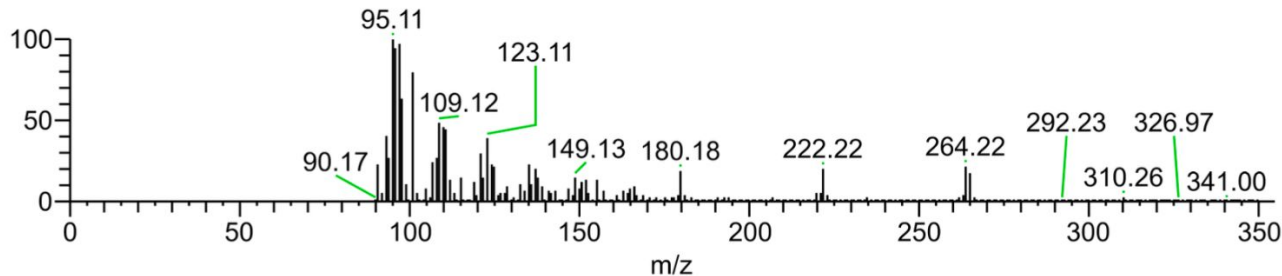
1465-1376

CH₂ and CH₃ bending

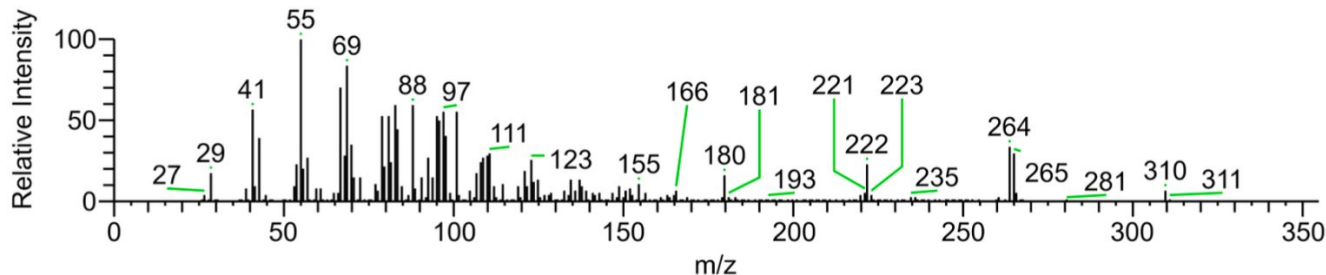
1174, 1163

C-O stretch from esters

Mass Spectrometry



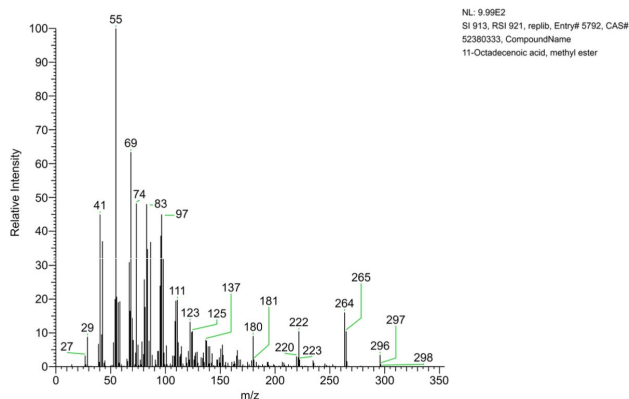
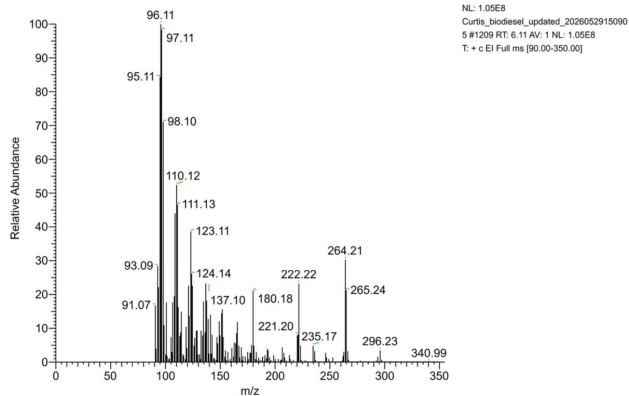
NL: 1.27E8
 Curtis_biodiesel_updated_2026052915090
 5 #1246 RT: 6.23 AV: 1 NL: 1.27E8
 T: + c EI Full ms [90.00-350.00]



NL: 9.99E2
 SI 914, RSI 922, replib, Entry# 5783, CAS#
 111626, CompoundName Ethyl Oleate

Hit	SI	RSI	Prob.	Molecular Wei...	Chemical For...	Name	Library Name	NIST Record...
1	914	922	38.93	310.28718	C20H38O2	Ethyl Oleate	replib	413911
2	856	868	38.93	310.28718	C20H38O2	Ethyl Oleate (E)-9-	replib	150151

Mass Spectrometry

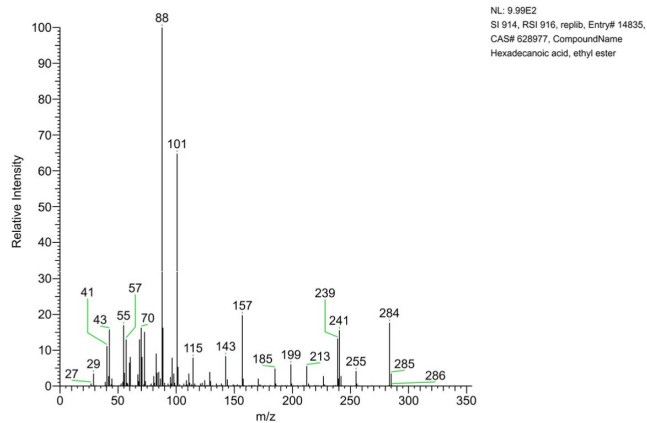
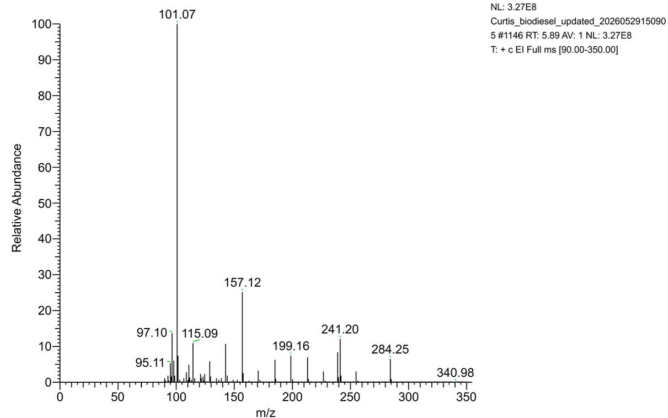


Hit	SI	RSI	Prob.	Molecular Wei...	Chemical For...	Name	Library Name	NIST Record...
1	913	921	12.15	296.27153	C19H36O2	"11-Octadecenoic acid, methyl ester"	replib	333556
2	909	917	10.26	296.27153	C19H36O2	"9-Octadecenoic acid, methyl ester, (E)-"	replib	333222
3	906	914	9.07	296.27153	C19H36O2	"trans-13-Octadecenoic acid, methyl ester"	mainlib	333613
4	902	910	7.66	296.27153	C19H36O2	"cis-13-Octadecenoic acid, methyl ester"	mainlib	333583
5	901	909	7.36	296.27153	C19H36O2	"6-Octadecenoic acid, methyl ester, (Z)-"	mainlib	333217
						"9-		

Compound Name:11-Octadecenoic acid, methyl ester
Chemical Name:C19H36O2



Mass Spectrometry

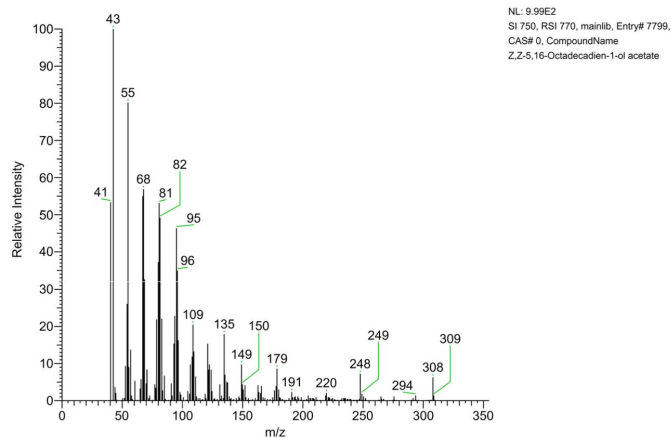
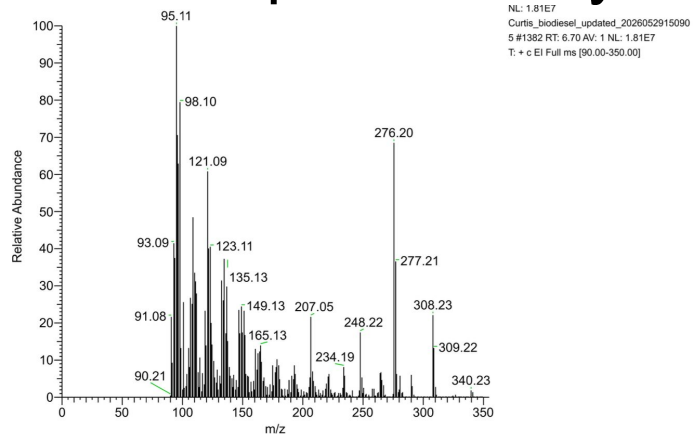


Hit	SI	RSI	Prob.	Molecular Wei...	Chemical For...	Name	Library Name	NIST Record...
1	914	916	93.4	284.27153	C18H36O2	"Hexadecanoic acid, ethyl ester"	replib	414789
2	904	906	93.4	284.27153	C18H36O2	"Hexadecanoic acid, ethyl ester"	mainlib	233204
3	851	923	93.4	284.27153	C18H36O2	"Hexadecanoic acid, ethyl ester"	replib	249073
4	789	844	4.2	284.27153	C18H36O2	"Hexadecanoic acid, 2-methyl-, methyl ester"	mainlib	68773

Compound Name: Hexadecanoic acid, ethyl ester
Chemical Name: C18H36O2



Mass Spectrometry



Hit	SI	RSI	Prob.	Molecular Wei...	Chemical For...	Name	Library Name	NIST Record...
1	750	770	25.17	308.27153	C20H36O2	"Z,Z-5,16-Octadecadien-1-ol acetate"	mainlib	130958
2	747	776	22.24	294.25588	C19H34O2	"E,E,Z-1,3,12-Nonadecatriene-5,14-diol"	mainlib	131114
3	740	835	17.03	280.276615	C19H36O	"2-Methyl-E,E-3,13-octadecadien-1-ol"	mainlib	130906
						"Pyrano[4,3-b]benzopyran-1,9-dione, 5a-		

Compound Name:Z,Z-5,16-Octadecadien-1-ol acetate
Chemical Name:C20H36O2



Thank You!